PORTFOLIO · 2026 · PROTOTYPE

Repair Reader

A manufacturer's repair manual goes in; a Georgian work card with steps, parts and tools comes out.



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AI prototype for a premium automotive service centre · manuals from two manufacturers

THE PROBLEM

Right facts, wrong language, wrong place

Written in English

Technicians work in Georgian, and a literal translation of a repair step reads like nonsense.

Spread across the document

Tools and technical values sit in tables at the start. Steps only point back at them.

What must be renewed?

Seals and self-locking nuts are single-use; other parts only if worn. Missing one means a second visit.

HOW IT WORKS

One manual, one card

01

Advisor

Drop the PDF

Any repair instruction, typed or scanned.

02

Advisor

Enter labour time

The manuals do not carry it, so a person enters it instead of the model guessing.

03

AI

Extract and translate

Every field of a fixed form: steps, warnings, parts, tools, fluids.

04

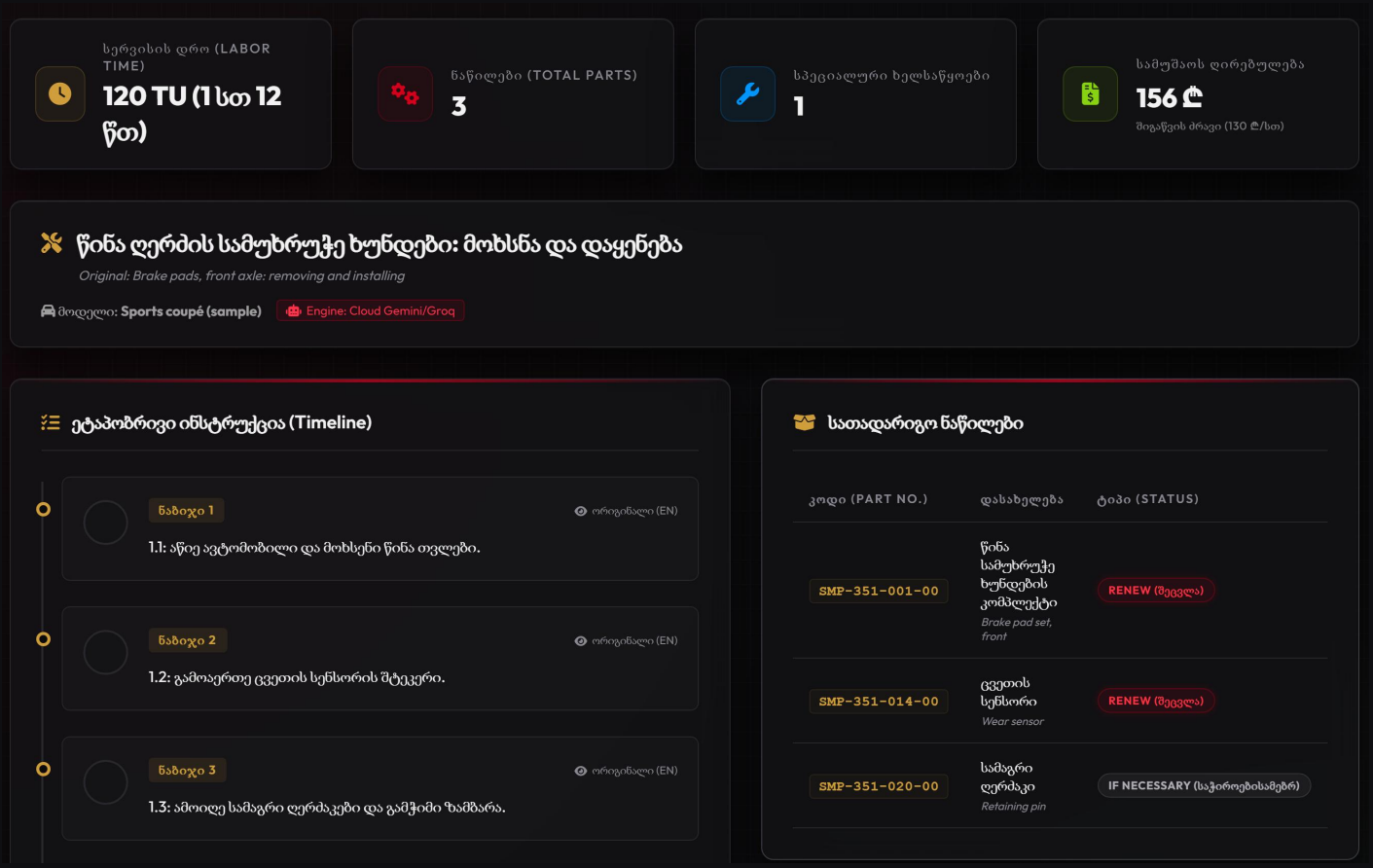
Workshop

Print the work card

One page per job for the technician, with the steps kept whole across page breaks.

THE RESULT

Answers before details



The advisor's four numbers

Labour time, part count, special tools and a labour cost estimate, at the top.

Title in both languages

Georgian first, the original underneath, with the model it applies to.

Cached by content

The same PDF is never analysed twice: its fingerprint finds the saved result.

Sample data · not a repair specification

THE RESULT

Always traceable

Original under translation

Each Georgian step, warning and part name keeps the English text beneath it.

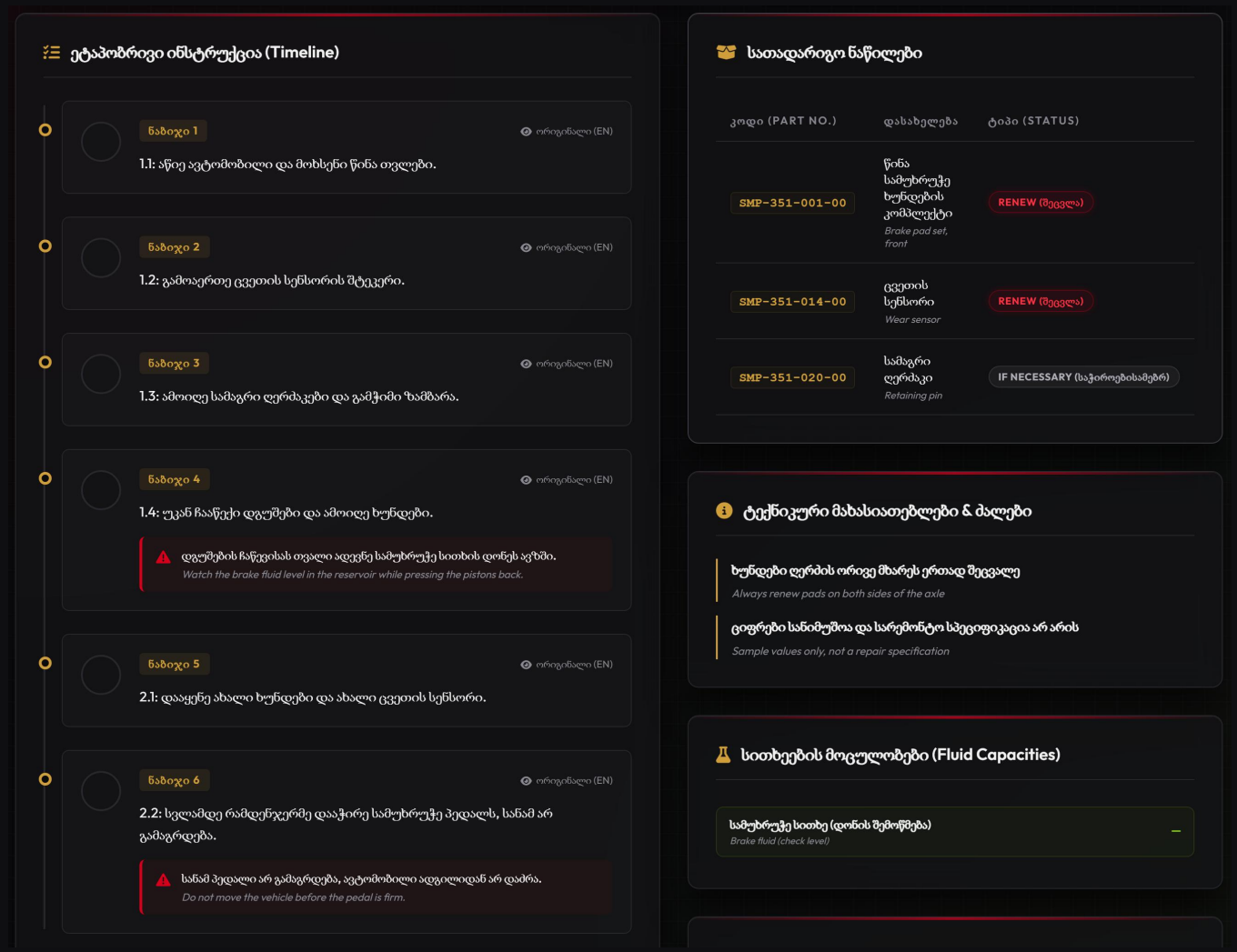
Warnings stay on their step

A caution belongs to the step it guards, not to a list at the end.

Renew or if necessary

Single-use parts are marked apart from parts replaced only on wear.

Sample data · not a repair specification



THE KEY DECISION

Ask the model to fill in a form, not to write an answer.

10

required fields in the schema

2,106

glossary terms in the prompt

Structured output means every result has steps, parts, tools and fluids in the same place, or it fails loudly. The glossary, including workshop slang, keeps the translation in the words technicians actually use.

HOW IT IS BUILT

Reading a PDF properly

Page by page, text or scan

Each page is classified. Scanned pages go to the vision model at 200 DPI, text pages stay cheap text.

Skip what is not repair

Covers, contents pages and legal notices are dropped before the model sees them.

Three ways to get an answer

Rotating keys on the main model, a second provider behind it, and a local model as the last resort.

Technical tables mapped back

Tools and values from the first-page tables are attached to the steps that reference them.

HOW IT IS BUILT

The architecture

Browser

Static HTML, CSS and JavaScript. A voice assistant answers part questions in Georgian.

API

Python and FastAPI in Docker, deployed by a GitHub Action. Rate-limited per client.

Models and storage

Gemini 2.5 Flash with structured output, Llama 3.3 70B as backup, results cached in cloud storage.

200 DPI

page images for scanned manuals

SHA-256

fingerprint: one analysis per manual

130

commits

WHAT I LEARNED

What the prototype taught me

Check every figure

A torque value from a model is a claim, not a spec. It must be traced to the source before a technician uses it. That became its own review step.

Don't let the model fill gaps

Asked for labour time the manual lacks, the model estimated. Now the app asks a person first, and their figure replaces the guess.

Scans hide in text PDFs

A document can read as text overall and still have scanned pages that silently vanish. Routing per page fixed it.

SUMMARY

The project in numbers

2,106

glossary terms, slang included

10

required fields per result

3

model tiers: main, backup, local

1

analysis per manual, ever

Python · FastAPI · PyMuPDF · Gemini structured output · Groq · Supabase Storage · Docker

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